Subjective Judging and the Host Effect at Japan's National Sports Festival: A 47-Prefecture Panel Analysis with a 2024-2025 Cross-Sectional Interaction Test

Running title: Subjective judging and the Kokutai host effect

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Abstract

Background: Host-country advantage in international sport is well established, but Balmer, Nevill & Williams (2003) documented that the advantage is concentrated in *subjectively judged* events (boxing, gymnastics) rather than in objectively measured events (track, weightlifting). Japan's National Sports Festival (Kokutai) is an unusually clean natural experiment for this hypothesis — 47 prefectures rotating as hosts across 80 editions since 1946, with essentially identical eligibility rules — yet the leading Japanese-language predecessor (Funahashi et al. 2016) documented a robust host bonus without disaggregating by sport type. We test whether the Kokutai host bonus is larger in subjectively judged sports than in objectively measured ones.

Methods: We assembled a 47-prefecture panel of Kokutai overall rankings spanning 2012-2022 (n = 423 prefecture-years, main analysis) and reproduced Funahashi et al. (2016)'s 2003-2011 specification (n = 423) for direct comparability. As primary specifications we fitted (i) a pooled ordered logit on 1-8 finishing ranks censored at "outside top 8" with log-population and log-GDP as covariates (no prefecture or year fixed effects), (ii) a binary logit on top-1 (championship) status with the same covariate set, and (iii) a Csurilla-style staged analysis (host only → +population → +GDP → +prefecture fixed effects (FE) → +year FE) — under which the two primary specifications above are numerically identical to the Csurilla M3 (+ log GDP) row, an equivalence we make explicit below. Because sport-by-prefecture-by-year score data are only publicly available for the two most recent editions (2024 Saga and 2025 Shiga), we tested the objective/subjective interaction in a stacked cross-section of 6,991 prefecture-sport-year cells (47 prefectures × 40-41 sports × 2 years × 2 trophies, prefecture-clustered SE). Sports were classified as objective (n = 17), subjective (n = 11), or semi-subjective/team (n = 13) following Balmer, Nevill & Williams (2003).

Results: In the 2012-2022 panel, host prefectures took the championship in 6 of 9 host-years (66.7%) and finished in the top 8 in 9 of 9 (100%) — a complete separation on top-8 status that was handled descriptively. The pooled ordered logit (with log-population and log-GDP controls) yielded a host coefficient of -13.73 (SE = 2.03, prefecture-clustered; $p < 0.001$; negative meaning better rank), and the pooled binary logit for top-1 (same controls) gave +9.09 (SE = 2.13, prefecture-clustered; $p < 0.001$). Our replication of Funahashi et al. (2016) for 2003-2011 recovered a host coefficient of +1,575.45 in total score (SE = 57.58; $p < 0.001$; $R^2 = 0.94$), within 5.9% of Funahashi's reported +1,674.65 despite our specification omitting Funahashi's socioeconomic controls (ESRI 2008 System of National Accounts (2008SNA) data are not available before 2011). Csurilla-style staged confounder controls did *not* attenuate the host coefficient; on the contrary, adding population and GDP strengthened it — the inverse of the ~45% attenuation Csurilla & Fertő (2023) report for the Olympics, and consistent with a host effect that is not confounded by prefecture size. In the 2024-2025 cross-section, the primary specification (obj-vs-subj pure comparison, n = 4,744; the 13 semi-subjective/team sports excluded to isolate a pure objective-versus-subjective contrast so that β_{HS} captures the intended subjective-vs-objective contrast without blending a semi-subjective mass into the reference base) yielded a host main effect of +11.18 points per sport (SE = 1.70; $p < 0.001$) and a **host × subjective interaction of +20.27 points (SE = 9.15; cluster-robust $p = 0.027$)** — the marginal host bonus in subjectively judged sports was roughly *triple* the bonus in objectively measured sports. The inclusive sensitivity specification (n = 6,991, retaining all sports) yielded a qualitatively identical pattern with a smaller interaction point estimate (host main effect +16.60, interaction +16.68, cluster-robust $p = 0.031$), as expected from the semi-subjective mass that the primary specification excludes from the reference base. Because the cross-section contains only two treated prefectures (Saga 2024, Shiga 2025) out of 47, we complement each cluster-robust p-value with a wild-cluster bootstrap (Rademacher weights, restricted null, B = 999), which yielded $p = 0.175$ (primary) and $p = 0.156$ (inclusive); neither interaction attains conventional significance under this few-treated-clusters correction and both should be read as first empirical anchors rather than as decisive rejections of the null. Descriptively, the raw host-versus-nonhost gap was +37.9 points in subjective sports, +26.2 in semi-subjective/team, and +17.6 in objective — monotonically ordered as predicted by Balmer et al. (2003).

Conclusions: The Kokutai host bonus appears directionally non-uniform across sport types. The subjective-versus-objective decomposition, absent from prior Japanese work, shows a Balmer-2003-consistent direction: the results are directionally consistent with a larger host bonus in subjectively judged sports (whose outcomes depend on judge scoring) than in objectively measured ones (whose outcomes depend on a stopwatch or a tape measure), with a point estimate roughly 3× under the obj-vs-subj-pure primary specification and roughly 2× under the inclusive sensitivity specification retaining all sports. Because we can only estimate the interaction from two cross-sectional years (2 treated prefectures out of 47), the point estimate should be viewed as a first empirical anchor rather than a definitive magnitude, and the wild-cluster bootstrap that accounts for the few-treated-clusters structure does not reach conventional significance under either specification (primary $p = 0.175$; inclusive sensitivity $p = 0.156$). Nevertheless, the descriptive host-boost gap (+37.9 subjective vs. +17.6 objective, monotonically ordered as Balmer et al. (2003) predicts) and the log-outcome robustness check (interaction coefficient +0.293, same direction

as the linear specification) both support the Balmer et al. (2003) direction, and the finding closes a specific gap left by Funahashi et al. (2016) and provides Japanese quantitative evidence complementing the qualitative critiques of Suetsugu (2024, 2025).

Keywords: Home advantage, host effect, subjective judging, National Sports Festival, Kokutai, panel data, Japan, ordered logit, Balmer 2003

Introduction

Does hosting a sporting event genuinely improve the host's performance, and if so, through what mechanism? At the international scale, the answer is unambiguously yes: host countries reliably win more medals than their non-host baseline predicts, an effect documented across a century of Summer Olympic Games (Balmer, Nevill & Williams 2003)[3] and the Winter Olympic series (Balmer, Nevill & Williams 2001)[1], with a broader Olympic/Paralympic replication (Wilson & Ramchandani 2021)[2] over the modern 1988-2018 window. The mechanism, however, is contested. Six candidate pathways are typically invoked — host-country entry quotas, training-venue familiarity, home-crowd support, budget uplift, jet-lag/climate advantage, and, most contentiously, **subjective-judging bias in favor of the host** — but no single mechanism has been uniquely identified in any single dataset.

Balmer, Nevill & Williams (2003) proposed a diagnostic decomposition.[3] If home advantage were driven by facility familiarity, budget, or crowd support, it should appear roughly uniformly across all Olympic sports. If, on the other hand, referee or judge bias were a load-bearing mechanism, home advantage should be concentrated in sports where the outcome depends on human scoring rather than an objective clock or tape measure. Comparing five event families across a century of summer Olympic data, Balmer et al. (2003) found that home advantage is indeed disproportionately concentrated in subjectively judged sports — boxing and gymnastics, in the specific event groups reported in the paper's abstract — with much smaller or absent effects in objectively measured sports such as athletics and weightlifting. This decomposition provides an implicit test of the judging-bias mechanism that does not require access to individual judge scores.

Japan's National Sports Festival (国民体育大会, *Kokutai*; renamed 国民スポーツ大会 in 2024) offers an unusually clean setting for a Balmer-2003-style test. Since 1946, the Kokutai has rotated across all 47 prefectures on a fixed schedule, with essentially identical eligibility rules and a stable overall-score system. The event is large (approximately 40 sports across a summer meet and a smaller winter meet), long-running (80 editions through 2026), and centrally administered by the Japan Sport Association (JSPO). Two headline findings are widely known within Japan. First, host prefectures win at a startling rate: from 1978 through 2015, the host prefecture took the championship trophy in **97.3%** of editions (36 of 37 non-cancelled meets). Second, in a distinct pattern beginning in 2016, the host championship rate has collapsed to **37.5%** for 2016-2025 (3 of 8) — a discontinuity that has generated both journalistic speculation and one scholarly critique (Suetsugu, 2024, 2025)[4,5] that host outcomes reflect a "host-victory-first mindset" (a phrase corresponding to the "開催地優勝至上主義" framing that appears in the title of Suetsugu 2025[5]). (For the historical record, the popular phrase "*Nara hantei*" — a slur for perceived hometown scoring bias — refers to a 2018 boxing scandal centered on Yamane Akira, then president of the Japan Amateur Boxing Federation, as documented in Japanese national news coverage of the federation's July 2018 self-dissolution and subsequent JOC/JSPO governance reforms; the phrase does not refer to the Kokutai kendo competition as is sometimes claimed.)

The most substantial quantitative predecessor is Funahashi, Hibino, Ishiguro & Mano (2016).[6] They assembled a balanced 47-prefecture × 9-year panel (n = 423) for 2003-2011, regressed the male-female combined competition score on a full set of host-year event-time dummies (t-7 through t+7), and estimated a host-year bonus of +1,674.65 total-score points on top of prefecture and year fixed effects (R^2 = 0.86). Funahashi (2016)'s Table 4 lists the six candidate mechanisms invoked above and cites Balmer et al. (2001)[1] — Balmer's Winter Olympics paper — as their reference for the subjective-judging pathway. But Funahashi et al. did **not** cite Balmer et al. (2003), and did not include either a sport-type indicator or a host × sport-type interaction in any of their specifications. The subjective-versus-objective decomposition that Balmer et al. (2003) formalized is therefore absent from the Japanese panel literature. Chiba's earlier qualitative treatment (1987)[7] likewise enumerates five host-victory mechanisms — the full-entry system, alleged combination-drawing bias, transferred-athlete contributions, intra-prefecture athlete development, and prefecture-wide civic support — and explicitly does not mention refereeing or judging bias in any of the five.

Institutionally, the Japanese Kokutai stands in mild contrast to the Korean equivalent. The Korean Sport & Olympic Committee's national comprehensive sports competition regulations explicitly award a **20% bonus** to the host city's overall score, codified from 2010 onward (previously 10% from 2001).[8] Japan has no such explicit host bonus in its scoring rules — the JSPO regulations we examined contain no analogous provision — which makes the Japanese host advantage, if present, a purely emergent phenomenon rather than an institutional artifact. This makes the Balmer et al. (2003)-style objective/subjective decomposition especially informative in the Japanese case: an interaction that survives in the absence of an institutional bonus is more strongly suggestive of the crowd-and-judging pathway.

We accordingly ask three questions. First (**replication**), does a Kokutai host bonus of a similar order of magnitude survive an extension of Funahashi's 2003-2011 panel through 2022, and does our reproduction of Funahashi's specification recover his headline coefficient? Second (**structural change**), how should the discontinuity in host-victory rate around 2016 be characterized — as a single break, or as two clusters (2016-17 and 2022-24) separated by a return to host wins in 2018-19? Third and centrally (**novel test**), is the Kokutai host advantage larger in subjectively judged sports than in objectively measured ones, as Balmer et al. (2003) would predict? Our answers, in brief, are (i) yes: the host coefficient replicates within ~6% of Funahashi's estimate; (ii) two clusters, not one, requiring a two-layer event-study design; and (iii) yes in direction and magnitude: the host × subjective interaction is +20.27 points on a base host effect of +11.18 in the primary obj-vs-subj-pure specification (cluster-robust $p = 0.027$; wild-cluster bootstrap $p = 0.175$ under the few-treated-clusters correction), directionally consistent with a larger host bonus in subjective sports than in objective ones (point estimate roughly 3× the objective bonus; the inclusive sensitivity specification retaining all sports yields +16.68 on a base of +16.60, cluster-robust p = 0.031, bootstrap p = 0.156) — with the descriptive host-boost gap (+37.9 subjective vs. +17.6 objective) monotonically ordered as Balmer et al. (2003) predicts, though the statistical significance of the point estimate should be read as tentative under the two-treated-prefecture design.

Methods

Data sources

The main analysis panel was assembled from three complementary sources. Overall prefecture rankings for the emperor's cup (天皇杯; male-female combined) and empress's cup (皇后杯; female only) from the 3rd through 79th editions were parsed from the Nagano Prefecture Sports Association ranked-listings page (https://www.nagano-sports.or.jp/kokutai/record/high_rank.html), which reports the top 8 finishers per edition per trophy. Where an edition-specific total-score PDF was available on the JSPO official archive (https://www.japan-sports.or.jp/kokutai/tabid183.html), the individual-edition PDF was parsed with `pdfplumber` (Python 3.14) to recover per-prefecture total scores; the machine-readable PDFs covered editions 58-67 (2003-2012) reliably, while editions 68-77 (2013-2022) were either 404 responses or image-only PDFs from which text could not be extracted (see Limitations). For the 2024 Saga (78th) and 2025 Shiga (79th) editions, JSPO published Excel files with the full 47-prefecture × 37-sport breakdown (`78_score_data.xls`, `78_score_all_data.xls`, `79_score_all_data.xls`, `79_score_deta.xls`); these were parsed with `python-calamine` (the Rust-based `xlrd` replacement, required because both files use a formula function ID that `xlrd` misidentifies).

Sport classification followed Balmer, Nevill & Williams (2003).[3] Of the 40-41 Kokutai sports (37 summer + 3 winter, plus clay-target shooting in the 2024 Saga edition only), **17** were classified as **objective** (client-independent measurement: track and field, swimming, cycling, rowing, canoe, weightlifting, sailing, ski, skating, archery, kyudo, rifle shooting, clay-target shooting, golf, bowling, triathlon, sport climbing; clay-target shooting was appended in the 2024 edition and removed in 2025, so the effective objective count is 17 in 2024 and 16 in 2025), 11 as **subjective** (referee/judge scoring: kendo, judo, gymnastics, karate, jukendo, naginata, boxing, fencing, wrestling, sumo, equestrian), and 13 as **semi-subjective / team** (team ball sports with referee-mediated but primarily objective outcomes: soccer, basketball, volleyball, handball, hockey, rugby, softball, baseball, badminton, table tennis, soft tennis, tennis, ice hockey). The classification is implemented in `src/sport_classifier.py` and unit-tested against the Balmer et al. (2003) decision rules.

Two socioeconomic covariates were assembled from the Cabinet Office's Economic and Social Research Institute (ESRI) prefectural accounts (令和4年度版; https://www.esri.cao.go.jp/jp/sna/data/data_list/kenmin/files/contents/main_2022.html): total population (`soukatu9`) and nominal prefectural GDP (`soukatu1`), both on the 2008SNA basis covering fiscal 2011-2022 for all 47 prefectures (a subsequent 令和5年度版 partial release covering 31 prefectures and 2 designated cities has been made available on the ESRI page, but complete 47-prefecture R5 data are not yet published; the 2011-2022 window is therefore retained). Log-transformed versions were merged onto the panel via `merge_confounders()`; coverage is 100% (1,128 of 1,128) for non-special-edition cells in 2011-2022 and 100% NaN outside that window, restricting the covariate-adjusted analyses to 2012-2022.

Study design

The primary analysis is organized around three complementary specifications.

1. Main panel (2012-2022, n = 423 prefecture-years): Pooled ordered logit on rank (1-8, with "outside top 8" coded as rank 9) and pooled binary logit on top-1 (championship) status. Both pooled specifications include log-population and log-GDP as covariates by construction; the label "pooled" denotes the absence of prefecture and year fixed effects (FE), not the absence of covariates. The two primary pooled coefficients reported in Table 2 are therefore numerically identical to the corresponding M3 (+ log GDP) rows of the Csurilla-style staged specification (Table 4). The corresponding prefecture-and-year fixed-effect versions were fitted for comparability with Funahashi's specification, but with fixed effects the small-sample maximum likelihood exhibited the expected complete-separation instability on top-1 (host prefectures win their host year almost deterministically, and Tokyo wins the majority of non-host years); we report the pooled specifications as primary and the FE specifications as sensitivity checks (with the FE specifications' inflated point estimates reported honestly and interpreted as fingerprints of separation rather than as substantively estimable effects). Standard errors for the pooled specifications are prefecture-clustered (47 clusters, Liang-Zeger sandwich estimator) to account for repeated observations of each prefecture across the 9 years of the panel; the fixed-effect specifications retain analytical (Fisher-information) SEs because prefecture FE renders further prefecture-clustering mechanically redundant, and their degenerate SEs are flagged as non-inferential regardless. The proportional-odds (PO) assumption underlying the ordered logit was probed with a Brant-style partial-PO diagnostic (each rank threshold fit as a separate prefecture-clustered binary logit, with the is_host coefficient compared across thresholds via an inverse-variance-weighted chi-square); the diagnostic and its tractability limitations on this dataset are reported in Limitations.

2. Funahashi (2016) replication (2003-2011, n = 423): Ordinary least squares (OLS) on total emperor's-cup score with prefecture and year fixed effects and a single host-year dummy — Funahashi (2016)'s Table 4 "base" specification, but omitting Funahashi's population/GDP/headquarters/transfer-athlete/participants controls because ESRI 2008SNA GDP series do not extend before 2011. A pooled OLS without fixed effects and a +1-year-extended specification (2003-2012) are reported as sensitivity checks. The prefecture-clustered standard errors used for statistical inference are computed on 47 clusters. The purpose is validation of the pipeline against a published estimate, not a novel estimate.

3. 2024-2025 cross-section for the subjective-objective interaction (n = 6,991): Because sport-by-prefecture score data are only publicly released for the 2024 and 2025 editions, the Balmer et al. (2003)-style host × sport-type interaction is estimated on a stacked cross-section. The theoretical panel size (a full cartesian product of prefectures and sport-year-trophy cells) is 47 prefectures × ([40 tennou + 35 kougou] in 2024 Saga + [40 tennou + 36 kougou] in 2025 Shiga) = 47 × 151 = **7,097 cells**; the regression frame retains **6,991 cells** (a 106-cell, 1.5% shortfall) after handling of the clay-target-shooting/boxing swap between the two editions and the drop of unbalanced cells for prefectures that did not participate in specific sports and are consequently absent from the JSPO overall-standings publication for that sport-year. The specification is $\text{score}_{isc} = \alpha + \beta_H \cdot \text{host}_{isc} + \beta_S \cdot \text{subj}_c + \beta_{HS} \cdot (\text{host} \times \text{subj})_{isc} + \eta_i + \kappa_c + \nu_t + \xi_p + \epsilon_{isc}$ where η_i is a prefecture fixed effect, κ_c is a sport fixed effect, ν_t is a year fixed effect, and ξ_p is a trophy (tennou/kougou) fixed effect (the trophy fixed effect is included to control for the systematically different competition-count and scoring structures between the emperor's cup and the empress cup). Prefecture-clustered robust standard errors are used (47 clusters). The primary specification excludes the 13 semi-subjective/team sports so that β_HS captures the pure obj-vs-subj contrast (subjective sports vs. objectively measured reference), yielding an estimation sample of n = 4,744 obj + subj cells across 28 sports; this eliminates the mixing bias that would otherwise fold semi-subjective sports into the reference base. As an inclusive sensitivity check, the same equation is fitted on the full n = 6,991 sample retaining all sports; this reproduces what a mechanical application of the Balmer et al. (2003) 3-category taxonomy without exclusion would yield, at the cost of the mixed obj + semi reference base. A three-way interaction fully separating objective, semi-subjective, and subjective was also fitted on the inclusive sample but produced a cluster-robust standard error that was numerically undefined (the cluster meat matrix loses rank under (k = 94 parameters) / (G = 47 clusters) ≈ 2, the few-clusters regime of Kezdi 2004), so we report the three-way specification diagnostically only (point estimates only, no inference). Similarly, a sport × trophy interaction augmentation of the primary specification (adding sport × cup fixed effects to the primary spec so that per-sport tennou-vs-kougou structural differences are absorbed as an additional diagnostic on whether within-sport tennou-vs-kougou differences confound β_HS) was fitted; this too produced a numerically undefined cluster-robust standard error under the same few-clusters regime (k = 131 parameters / G = 47 clusters ≈ 2.8), so we report it diagnostically only as an additional point-estimate consistency check on β_HS. A log-outcome specification ($\log(1 + \text{score})$) on the inclusive sample is reported as a robustness check. Because absent prefecture-sport cells (106 of the theoretical 7,097; 1.5%) may represent non-participation rather than missing data, we additionally estimated a zero-imputed specification treating absent cells as zero score as a non-participation vs missing-data sensitivity check; the resulting spec is reported diagnostically only because the augmented cluster meat matrix produces a rank-deficient cluster-robust standard error, and we use it as a directional check on whether the primary interaction is driven by the missing-data handling convention rather than the underlying host-subjectivity structure.

Csurilla-style staged confounder controls

We fitted the sequence M1 (host only) → M2 (+ log population) → M3 (+ log GDP) → M4 (+ prefecture FE) → M5 (+ year FE), separately for ordered rank and top-1, on the 2012-2022 panel. Csurilla & Fertő (2023)[9] report that adding population and per-capita GDP to a host-country regression attenuates the Olympic host coefficient by approximately 45% (from ~0.47 to ~0.26 for total medals). Whether the same pattern holds at the Kokutai — where within-country socioeconomic variance is smaller than within-Olympic-country variance — is not obvious *ex ante*.

Event-study design (supplementary)

To decouple the pre-2005 *furusato*-athlete-rule regime from the post-2016 regime, a two-layer event-study was specified as a design check rather than an independent identifying analysis (linear-probability model on top-1 status, prefecture-clustered SE, formal parallel-trend Wald tests). Layer 1 contains the single 2002 Kochi pre-2005 host-loss event; Layer 2 stacks the five post-2016 host-loss events (2016 Iwate, 2017 Ehime, 2022 Tochigi, 2023 Kagoshima special, 2024 Saga). Because the treated group exhibits non-zero top-1 status essentially only in the host year, the identifying variation is thin; full design details, trails, and Wald test statistics are reported in Supplementary Materials §S1.

Statistical software

All analyses were performed with Python 3.14 using pandas 3.0.5, numpy 2.5.1, statsmodels 0.14.6, patsy 1.0.2, scipy 1.18.0, python-calamine 0.8.2, pdfplumber 0.11.10, and matplotlib 3.11.1. Prefecture-cluster-robust standard errors (47 clusters, Liang-Zeger sandwich) are reported for all main-panel pooled specifications, the Csurilla-style M1-M3 stages, the replication OLS, and the 2024-2025 cross-section; Fisher-information standard errors are retained for the M4/M5 fixed-effect specifications where prefecture FE renders further clustering mechanically redundant and where complete separation drives SEs to numerical degeneracy in any case. Statistical tests are two-sided at $\alpha = 0.05$. Unit tests ($n = 309$ total, all passing) verify the numerical integrity of each analysis module against small deterministic fixtures; the test suite is distributed across `sport_classifier` (51 tests), `analysis_cross_section_2024_2025` (51 tests, including zero-imputed sensitivity, Table 5d 4-variant wild-cluster bootstrap, and pivotal 95% CI), `data_loader` (75 tests covering JSPO xls parsing across the 78/79 editions), `analysis_main` (24 tests), `definitions` (25 tests), `panel_builder` (20 tests), `analysis_replication` (19 tests), `analysis_event_study` (18 tests), `analysis_confounders` (17 tests), and `plots` (9 tests).

Multiple comparisons. The staged analytic plan treats the 2024-2025 cross-section host × subjective interaction (β_HS, Table 5 primary row) as the single confirmatory test of the paper's central hypothesis (that the Kokutai host bonus is larger in subjectively judged sports than in objectively measured ones). The pooled main-panel host coefficients (Tables 2 and 4), the Funahashi replication (Table 3), the Csurilla-style staged confounder specifications (Table 4 and Table 4b), and the two-layer event-study (Supplementary Materials §S1) are treated as replication, sensitivity, and design-check specifications respectively — they are not independent confirmatory tests of the central hypothesis and do not each require their own α-level adjustment. Consequently, we do not apply Bonferroni or Benjamini-Hochberg correction across the roughly 15 reported p-values in the manuscript. The confirmatory hypothesis is tested exactly once, and its outcome under the appropriate few-treated-clusters correction (wild-cluster bootstrap $p = 0.175$ primary, 0.156 inclusive sensitivity; see the wild-cluster bootstrap paragraphs in

Results and Limitations) is reported directly. A reader preferring an explicit Bonferroni-style guardrail may note that under even a very generous 5-test correction ($\alpha\_family = 0.05 \rightarrow \alpha\_single = 0.01$), the naive cluster-robust p (primary: 0.027; inclusive sensitivity: 0.031) would not remain significant, and the wild-cluster bootstrap p (primary: 0.175; inclusive: 0.156) would remain non-significant under either the uncorrected or the corrected threshold. The qualitative reading is therefore unchanged: the confirmatory interaction is significant under the uncorrected cluster-robust test only, and does not survive either the few-treated-clusters correction or a generous multiple-comparisons correction.

Ethical considerations

This study analyzes publicly available Kokutai overall-standings data released by the Japan Sport Association and prefectural sport associations. No individual-level, health, or personally identifying data are used, and no human subjects were recruited or contacted. Because the analysis is restricted to already-public aggregate competition results without any personally identifiable information, it does not meet the standard threshold for human-subjects research under either domestic (e.g., MEXT / MHLW — the Ministry of Education, Culture, Sports, Science and Technology and the Ministry of Health, Labour and Welfare — ethical guidelines for research using publicly available secondary data) or international (e.g., 45 CFR §46.102 for non-human-subjects data) frameworks; accordingly, no institutional review board approval or waiver was sought or required.

Results

Descriptive statistics (2012-2022 panel)

Across 2012-2022, the emperor's-cup panel contains 423 prefecture-year observations (47 prefectures × 9 non-special, non-cancelled years; see Table 1 for the descriptive summary). Host prefectures (n = 9 host-years) took the championship in 6 of 9 host-years (66.7%) — a rate two orders of magnitude above the non-host per-year top-1 rate of 3 in 414 (0.72%) — and appeared in the top 8 in **9 of 9 host-years (100%)**. This 100% top-8 rate constitutes a complete separation in a top-8 logit and was accordingly handled descriptively rather than as a fitted model: the fact of nine-for-nine top-8 host appearances is itself a stronger point estimate than any logit could produce on such a fixture. When the host prefecture did finish inside the top 8 (which is to say, always), the mean host rank was 1.33.

The three host-loss host-years within the 2012-2022 panel window (in which a non-host prefecture — in each case Tokyo — took the championship) were 2016 Iwate, 2017 Ehime, and 2022 Tochigi. Kagoshima's 2023 special edition (a COVID-delayed re-run of the cancelled 2020 program) and 2024 Saga also produced Tokyo championships outside the panel window, extending the run to 2022-24 in the descriptive time series. The six host-victory host-years within the panel window fall in 2012 Gifu, 2013 Tokyo (self-host), 2014 Nagasaki, 2015 Wakayama, 2018 Fukui, and 2019 Ibaraki (six wins in the six non-loss host-years within the panel window). The 2013 Tokyo cell is Tokyo hosting itself and is included in the "6 of 9" descriptive rate: Tokyo's host-year 1st-place finish (host_win = 1) is a legitimate host-win observation, notwithstanding Tokyo's generalized dominance in non-host years. As a sensitivity check, excluding the 2013 Tokyo cell yields 5 of 8 = 62.5%, still two orders of magnitude above the non-host per-year top-1 rate of 0.72% and in the same direction as the primary "6 of 9" rate.

Main panel: pooled ordered logit and top-1 logit (with log-population and log-GDP controls)

The pooled ordered logit on rank (censored at 9 = outside top 8), with log-population and log-GDP controls, yielded a host coefficient of **-13.73** (SE = 2.03, prefecture-clustered; $z = -6.76$; $p < 0.001$; n = 423), where a negative coefficient corresponds to better (numerically smaller) rank on the ordered scale. In log-odds terms this is an extraordinarily large shift — the median host prefecture-year sits at rank 1, versus a median non-host prefecture-year outside the top 8 — but the magnitude reflects the near-deterministic character of host top-8 status in the data rather than a subtle behavioral effect. The pooled binary logit on top-1 (same controls) gave a host coefficient of **+9.09** (SE = 2.13, prefecture-clustered; $z = 4.28$; **$p < 0.001$**, exact $p = 1.9 \times 10^{-5}$), corresponding to an odds ratio far above 100 that is best interpreted, again, as a fingerprint of the concentrated top-1 mass in the host-year cells rather than a stable behavioral log-odds. These two pooled coefficients are numerically identical to the M3 (+ log GDP) row of Table 4 and share the same prefecture-clustered SE; the "pooled" label denotes the absence of prefecture and year fixed effects only, not the absence of covariates or of clustering (see Methods §1). Under the naive Fisher-information SE that would ignore the repeated-observations structure of each prefecture across years, the corresponding SEs would be 1.54 for ordered rank and 2.80 for top-1 — the ordered-rank SE understates prefecture-level dependence while the top-1 SE overstates it, but neither correction changes the direction or the significance of the effect at $\alpha = 0.05$.

The prefecture-and-year fixed-effect versions, which more closely mirror Funahashi's specification, produced degenerate estimates as expected under complete or near-complete separation: the FE ordered logit gave a host coefficient of -49.56 (SE = 172.45; $p = 0.77$), a fingerprint of the optimizer's failure to identify a meaningful ordered-logit slope when the treated-versus-untreated cells within each prefecture and year are near-singleton; the FE top-1 logit gave a coefficient of +1,877 (SE = NaN), a classic separation blowup. These FE specifications are reported for transparency but should not be interpreted as coefficient estimates in the ordinary sense.

Replication of Funahashi et al. (2016), 2003-2011

Our reproduction of Funahashi's prefecture-and-year fixed-effect OLS on emperor's-cup total score for 2003-2011 (n = 423) yielded a host coefficient of **+1,575.45** total-score points (SE = 57.58, based on 47 prefecture clusters; $p < 0.001$; $R^2 = 0.94$). Funahashi (2016)'s Table 4 reports a corresponding base-specification coefficient of +1,674.65, so our reproduction lies within 5.9% of Funahashi's estimate. The residual gap is attributable to our omission of Funahashi's socioeconomic controls (population, GDP, headquarters count, transfer-athlete rule ratio, participant count), which we could not include because ESRI 2008SNA prefectural GDP series do not extend before fiscal 2011. Sensitivity checks corroborated the finding: a pooled OLS without fixed effects gave a coefficient of +1,733.29 (SE = 138.26), and a +1-year extension to 2003-2012 (n = 470) gave +1,604.36 (SE = 60.08). Together, the three specifications bracket Funahashi's headline value tightly. Figure 5 displays the three replication coefficients alongside Funahashi's +1,674.65 reference line, with 95% confidence intervals (1.96 × SE) shown as error bars. The `pooled_no_fe` specification lies within one SE of the reference, and the `funahashi_base` and `extended_2003_2012` specifications lie within 1.96 SE (95% CI) of the reference.

Csurilla-style staged confounder controls

The staged specification produced a *strengthening* rather than an attenuation of the host coefficient as socioeconomic controls were added, running in the opposite direction from the ~45% attenuation Csurilla & Fertő (2023) documented for Olympic host countries. This directional comparison is not a scale-comparable comparison: Csurilla & Fertő fit a zero-inflated negative-binomial model on total medal counts (log link, country-by-Olympics identifying variance), whereas our specifications are a binary logit on top-1 status and an ordered logit on rank (logit link, prefecture-by-Kokutai identifying variance). The absolute-percentage-change directions differ, but the two coefficients live on different scales and identify off different variance decompositions, so the comparison is best read as "the *direction* of the socioeconomic-controls effect on the host coefficient is inverted, on this paper's own scale" rather than as an IRR-to-IRR opposition. For the top-1 outcome, the M1 (host-only) coefficient was +5.61 (SE = 1.09, prefecture-clustered; $p < 0.001$); adding log population (M2) raised the coefficient to +10.90 (SE = 3.35; $p = 0.0011$); adding log GDP (M3) settled it at +9.09 (SE = 2.13; $p < 0.001$, exact $p = 1.9 \times 10^{-5}$). M4 (prefecture FE) and M5 (year FE) again produced separation-driven blowups (+498 and +1,877 respectively, with degenerate SEs; Fisher-information reported because prefecture FE renders clustering mechanically redundant) that are not interpretable as attenuation. For the ordered-rank outcome the same directional pattern held: M1 = -6.11 (SE = 1.21), M2 = -13.17 (SE = 2.01), M3 = -13.73 (SE = 2.03), all $p < 0.001$; a 2.25× *increase* in absolute magnitude from M1 to M3. Figure 4 displays the M1→M3 trail for the top-1 outcome, showing an anti-attenuation trajectory rather than the Csurilla-style ~45% decay. The most natural interpretation is that population and GDP are *negatively* correlated with the host coefficient's identifying variation at the Kokutai — Tokyo, an outlier on both covariates, is a large-population high-GDP prefecture that dominates non-host years, so controlling for population and GDP absorbs some of the non-host championship mass and (in relative terms) makes the host coefficient look larger. This is the opposite fingerprint from an Olympic setting in which the host is typically the largest and richest country in a given year: at the Kokutai, host prefectures are essentially random draws from the size distribution, so socioeconomic controls do not attenuate the host coefficient.

Event-study, two-layer design (supplementary)

The two-layer event-study parallel-trend Wald tests were formally non-rejecting in both layers (Wald $p = 1.0$), but the identifying variation is thin because the treated group exhibits non-zero top-1 status essentially only in the host year itself. The event-study point estimates are therefore near-zero by construction and should be read as a design check rather than as an independent identifying analysis of the shock timing. Full trails, Wald test statistics, and Supplementary Figure S1 are reported in Supplementary Materials §S1.

2024-2025 cross-section: the subjective-objective interaction

The stacked 2024-2025 cross-section (theoretical n = 6,991 prefecture-sport-year-trophy cells; 47 prefectures × 40-41 sports × 2 years × 2 trophies, after handling of the clay-target/boxing swap between the two editions) is the specification in which the Balmer et al. (2003)-style objective/subjective decomposition can be estimated. As the primary specification, we exclude the 13 semi-subjective/team sports to isolate a pure obj-vs-subj contrast: the resulting sample (n = 4,744 obj + subj cells; 28 sports; still 47 pref clusters), estimated with sport and year fixed effects and prefecture-clustered SEs, produces a host main effect of **+11.18 points per sport** (SE = 1.70; $p < 0.001$) and a **host × subjective interaction of +20.27 points** (SE = 9.15; cluster-robust **p = 0.027**). Because the main effect is a within-sport, cross-prefecture difference, +11.18 points is roughly the average per-sport score bonus a host prefecture receives in an *objectively measured* sport; +20.27 is the *additional* bonus in a subjective sport. Together, the model implies a total per-sport host bonus of roughly 31 points in a subjectively judged sport versus about 11 in an objectively measured one — directionally consistent with a larger host bonus in subjectively judged sports (point estimate roughly 3× the objective bonus). As a sensitivity check, retaining all 40-41 sports (the inclusive specification, n = 6,991) yields a host main effect of +16.60 (SE = 1.25; $p < 0.001$) and a host × subjective interaction of +16.68 (SE = 7.72; cluster-robust p = 0.031); the inclusive specification is qualitatively identical to the primary but yields a smaller interaction point estimate, as expected from the semi-subjective mass that the primary specification excludes from the reference base.

The 2-of-47 treated-cluster structure of this cross-section (only Saga in 2024 and Shiga in 2025 are host prefectures) exposes the cluster-robust variance formula to the Cameron & Miller (2015)[13] few-treated-clusters problem: with so few treated clusters, the cluster-robust standard error tends to be anti-conservative and the associated t-statistic is not well-approximated by its asymptotic distribution. We accordingly computed a wild-cluster bootstrap p-value (Cameron, Gelbach & Miller, 2008)[14] using Rademacher weights on prefecture clusters, a restricted null ($\beta_{HS} = 0$), and B = 999 iterations. The bootstrap p for the interaction under the primary specification was **p = 0.175** (observed t = 2.22), and under the inclusive specification p = 0.156 (observed t = 2.16); neither attains conventional (0.05) significance under this few-treated-clusters correction. The direction and magnitude of the interaction are preserved, but the point estimate should be read as a first empirical anchor at this stage of the data availability rather than as a decisive rejection of the null; the descriptive +37.9 vs. +17.6 subjective/objective gap and the log-outcome interaction coefficient (+0.293, same direction, cluster-robust p = 0.084) both support the Balmer et al. (2003) direction independently of the p-value scale.

The three-way specification with a semi-subjective interaction (fitted on the inclusive n = 6,991 sample) produced a cluster-robust standard error that was numerically undefined for both β_{HS} (+20.31) and $\beta_{H,semi}$ (+8.51) — a symptom of the (k = 94 parameters) / (G = 47 clusters) ≈ 2 few-clusters regime in which the cluster meat matrix loses rank (Kezdi 2004). We therefore report the three-way point estimates diagnostically only. The sport × trophy interaction augmentation of the primary specification (primary base with additional sport × cup fixed effects; adopted as a diagnostic on whether per-sport tennou-vs-kougou differences confound β_{HS}) likewise produced a cluster-robust standard error that was numerically undefined for β_HS under the same few-clusters regime (k = 131 parameters / G = 47 clusters ≈ 2.8; Kezdi 2004), yielding a β_HS point estimate of +21.97 that is reported diagnostically only. The three-way β_HS point estimate (+20.31) and the sport × trophy augmentation's β_HS point estimate (+21.97) are both notably close to the primary specification's +20.27, corroborating the direction and magnitude of the pure obj-vs-subj contrast that the primary specification is designed to isolate through four internally consistent point estimates. The log-outcome specification (inclusive sample) returned a host coefficient of +0.467 (SE = 0.057; $p < 0.001$) and an interaction coefficient of +0.293 (SE = 0.170; $p = 0.084$, marginal), directionally consistent with the linear specification. A zero-imputed sensitivity specification, treating the 106 absent prefecture-sport cells (1.5% of the theoretical 7,097 cartesian; adopted as a non-participation vs missing-data check) as score = 0 rather than as missing data, yields a β_HS point estimate of +19.86 (n = 4,841 primary-spec cells), within 2.04% of the primary specification's +20.27 (Table 5, zero-imputed row). The cluster-robust SE is numerically undefined under the augmented meat matrix rank-deficient regime, so we report the zero-imputed point estimate diagnostically only; the directional invariance of β_HS to the missing-data handling convention (primary +20.27 vs zero-imputed +19.86, direction (+) preserved) further supports that the primary interaction is not driven by the JSPO overall-standings publication's convention of dropping non-participating prefecture-sport cells.

Descriptively, the raw host-versus-nonhost per-sport score gap monotonically increased across the three sport categories in exactly the direction Balmer et al. (2003) predicts (Table 6; Figure 2): objective sports showed a gap of **+17.60 points** (host mean 43.63 vs. non-host mean 26.03, n_host = 65 cells), semi-subjective/team sports **+26.18 points** (47.03 vs. 20.85, n_host = 48), and subjective sports **+37.91 points** (57.24 vs. 19.33, n_host = 38). The subjective host boost of +37.9 is the largest of the three, and its gap over the objective +17.6 (a ratio of 2.15) is close to the model-implied doubling. Figure 2 displays the three category-specific host boosts with 95% confidence intervals.

Structural change: two clusters, not one

The raw host-championship rate declines from 97.3% for 1978-2015 (36 of 37 non-cancelled meets) to 37.5% for 2016-2025 (3 of 8 non-cancelled meets). This is consistent with a structural break around 2016. However, decomposing the 2016-2025 window reveals *two* clusters of host losses (2016-17 and 2022-24) separated by a full return to host wins in 2018 Fukui (host score 2,896 vs. Tokyo's 2,246) and 2019 Ibaraki (2,569 vs. 2,217). A single-break DID would blur these two clusters into one and misspecify the intermediate 2018-19 period. The two-layer event-study specification described in Methods was adopted precisely to preserve this structure. Figure 1 displays the emperor's-cup host championship-rate time series with the six host-loss years (2002 Kochi, 2016 Iwate, 2017 Ehime, 2022 Tochigi, 2023 Kagoshima special, 2024 Saga) marked; the two post-2016 clusters and the 2018-19 island are clearly visible.

Discussion

The Balmer et al. (2003) decomposition is directionally supported at the Kokutai

The central finding of this study is that the Kokutai host bonus appears to be directionally non-uniform across sport types: the marginal bonus in subjectively judged sports is directionally consistent with being larger than the bonus in objectively measured sports under the primary obj-vs-subj-pure specification (point estimate roughly 3× under primary; roughly 2× under the inclusive sensitivity specification retaining all sports). In the 2024-2025 stacked cross-section, the primary specification (semi-excluded, obj-vs-subj pure) yields a host main effect of +11.18 points per sport and a host × subjective interaction of +20.27 points (cluster-robust p = 0.027; wild-cluster bootstrap p = 0.175 under the few-treated-clusters correction), producing a model-implied total per-sport bonus of roughly 31 points in subjective sports against ~11 in objective ones — directionally consistent with a larger host bonus in subjective sports than in objective ones (point estimate roughly 3×). The inclusive sensitivity specification retaining all sports yields +16.60 base and +16.68 interaction (cluster-robust p = 0.031, bootstrap p = 0.156), qualitatively identical to the primary but with a smaller interaction point estimate as expected from the semi-subjective mass that the primary excludes from the reference base. The raw descriptive gap orders the three sport categories monotonically as +37.9 (subjective) > +26.2 (semi-subjective) > +17.6 (objective) — precisely the ordering that Balmer, Nevill & Williams (2003) documented at the summer Olympics. Given that the wild-cluster bootstrap does not reach conventional significance under the two-treated-cluster structure of this cross-section, we frame this as a *directionally* consistent first quantitative anchor for the Balmer et al. (2003) decomposition on Kokutai data — supported concordantly by (a) the raw descriptive monotonicity, (b) the log-outcome interaction direction, and (c) the linear interaction point estimate — rather than a decisive statistical confirmation. The Cameron-Gelbach-Miller (2008) pivotal wild-cluster bootstrap 95% confidence intervals ([-0.82, +41.36] under the primary specification; [-0.42, +33.86] under the inclusive sensitivity) further make the marginal-non-significance explicit at the interval level: both intervals cross zero by essentially the width of a rounding tolerance, consistent with the bootstrap p values reported in Table 5 (primary 0.175, inclusive 0.156) but preserving the substantial positive-mass concentration of the point estimate. It nevertheless closes the specific gap left by Funahashi, Hibino, Ishiguro & Mano (2016), whose 2003-2011 panel documented a robust overall host bonus but did not disaggregate by sport type.

A sport × trophy interaction augmentation of the primary specification, added as a diagnostic on whether emperor’s-cup vs empress-cup structural differences within sports confound the subjective host boost, yields a β_HS point estimate of +21.97 that is diagnostic-only under the k/G ≈ 2.8 few-clusters regime (cluster SE numerically undefined; Kezdi 2004). That the augmented point estimate falls within 9% of both the primary specification’s +20.27 and the three-way specification’s +20.31 provides a third independent point-estimate corroboration that the subjective host boost is not driven by trophy-level structural heterogeneity within sports; the tennou/kougou distinction absorbs essentially no substantive variance in β_HS. Together the primary, inclusive, three-way, and sport × trophy specifications form four internally consistent point estimates for β_HS across a plausible range of specification choices.

A three-variant classification sensitivity analysis, re-estimating the primary specification under alternative delimitations of the “subjective” category, further supports the invariance of β_HS to reasonable classification choices (Table 5d). The pure-judged variant (subjective narrowed to 体操 + 空手道 + なぎなた + 馬術, with the seven combat sports 剣道/柔道/銃剣道/ボクシング/フェンシング/レスリング/相撲 reclassified as objective) yields β_HS = +18.99 (SE = 17.71, p = 0.284), directionally consistent with the primary but with substantially inflated SE from the reduced subjective-host cell count. The combat-excluded variant, which drops all nine combat sports from the analysis and isolates pure artistic scoring (subjective = 体操 gymnastics + 馬術 equestrian only), yields β_HS = **+31.81** (SE = 8.20, cluster-robust **p < 0.001**; wild-cluster bootstrap p = 0.174, matching the primary’s bootstrap p = 0.175 within 0.001 under the few-treated-clusters correction applied consistently across all four variants) — the largest cluster-robust point estimate across all four variants, consistent with a mechanism in which host bias is strongest under pure artistic scoring rather than under combat-referee judgment; under the wild-cluster bootstrap correction, the no-combat variant’s inferential status is the same as the primary specification’s rather than uniquely stronger. The combat-to-semi variant (fencing/wrestling/sumo reclassified as semi-subjective, subjective = 8) yields β_HS = +26.40 (SE = 11.01, p = 0.016). All four variants preserve β_HS > 0 across the range +18.99 to +31.81, with the primary specification’s +20.27 sitting inside this range; the qualitative directional finding is therefore invariant to how the subjective category is delimited within the plausible range spanned by Balmer et al. (2003)-consistent classification schemes.

The finding is consistent with, but does not require, a judging-bias mechanism. The Balmer et al. (2003) decomposition is an *implicit* test — an interaction of the observed type is what a judging-bias mechanism would produce, but a purely crowd-effect mechanism that is stronger in enclosed, spectator-dense subjective-sport venues (typical: judo, kendo, gymnastics halls) than in open-air objective-sport venues (typical: track and field, cycling road courses) would produce the same pattern. Without a judge-level dataset we cannot uniquely identify judging bias as the mechanism. What we can say is that a purely uniform material mechanism would be less likely to produce this sport-type gradient, although sport-specific preparation, venue familiarity, and transfer-athlete channels cannot be ruled out. The interaction we recover is more consistent with a crowd-and-judging pathway than with a purely uniform material one. Nomura (2022)[10] provides collateral evidence from Japanese J1 football that home advantage disappeared during the reduced-occupancy 2020 COVID-19 season and — through multiple-group structural equation modeling — that the crowd’s influence on referees’ decisions vanished under the same conditions, further supporting the crowd–referee pathway.

Two host-loss clusters, not one

The popular framing of a “post-2016 Tokyo winning streak” is factually incorrect. Host prefectures won both 2018 Fukui (2,896 vs. Tokyo’s 2,246) and 2019 Ibaraki (2,569 vs. Tokyo’s 2,217) under exactly the same national scoring rules that produced the Tokyo wins in 2016 Iwate and 2017 Ehime. The 2020 and 2021 editions were cancelled for COVID-19, with the 2020 Kagoshima program re-scheduled as a special edition in 2023. The correct structural pattern is therefore two clusters of host losses (2016-17 and 2022-24) separated by a return to normal pattern, plus the single 2002 Kochi outlier in the pre-2005 regime. The two-layer event-study specification we adopt (see Supplementary Materials §S1) preserves this structure; a naive single-break DID at 2016 would produce a biased estimate of the structural change. This suggests a modeling recommendation for future Japanese work on the Kokutai host effect: 2016 should not be treated as a single structural break, and a two-cluster or event-study framing that preserves the 2018-19 return-to-host-wins interlude is likely to be more informative.

East-Asian institutional context

The Kokutai host bonus emerges *in the absence of* an explicit institutional host bonus in JSPO scoring rules (Table 7 summarizes the cross-national institutional context). This stands in contrast to the Korean Sport & Olympic Committee's national comprehensive sports competition, which explicitly adds a 20% score bonus to the host city (10% from 2001, raised to 20% in 2010),[8] and to the qualitative characterization Lan and Yu (2012)[11] provide of the Chinese National Games host effect, whose five listed characteristics — high efficacy, gradual increase, delayed effect, exclusivity, and double-edgedness — describe roughly the same phenomenon we observe empirically at the Kokutai but in more diffuse qualitative language. The present study is arguably an implicit decomposition of Lan & Yu's "exclusivity" and "double-edgedness" characteristics: exclusivity because the host bonus is concentrated in a subset of sports (subjectively judged), and double-edgedness because a large host bonus in a subjective sport increases both the host's expected medal count and the perceived legitimacy risk of that medal (see the Suetsugu 2024/2025 qualitative critique below).

Overlap and boundary with Suetsugu (2024, 2025)

The Suetsugu papers[4,5] provide a karate-specific qualitative critique of Kokutai outcomes, using the "host-victory-first mindset" framing (a phrase corresponding to the "開催地優勝至上主義" title of Suetsugu 2025) in a single-sport context. Our whole-sport panel-quantitative decomposition is complementary rather than overlapping: Suetsugu's papers focus on a single subjectively judged sport (karate) and characterize the outcome pattern in that sport in normative terms; our analysis provides the sport-level statistical evidence that the pattern Suetsugu describes in karate is indeed sport-type-specific (larger in subjectively judged sports than in objectively measured ones) rather than merely idiosyncratic to karate. The two contributions can stand side by side: Suetsugu characterizes the qualitative concern within a single sport; we provide quantitative evidence consistent with the same interaction pattern across all sports.

We deliberately avoid the normative language Suetsugu uses. The observed interaction is consistent with judging bias, but it is also consistent with subject-sport-specific crowd effects, subject-sport-specific practice-environment effects, and subject-sport-specific transfer-athlete effects. The present study does not adjudicate among these mechanisms; it provides quantitative evidence consistent with their collective effect being larger in subjectively judged sports than in objectively measured ones, by a factor of 2.00× (inclusive sensitivity) to 2.81× (primary specification).

The identified interaction is therefore a reduced-form estimate that lumps judge-bias, crowd-effect, practice-environment, and transfer-athlete channels; the absence of a JSPO judge-of-record roster (see Limitations, Second) prevents attribution of the observed effect to any single channel. A judging-bias mechanism, if present, would plausibly operate through host prefectures that supply larger judge cohorts for subjectively judged sports (gymnastics, kendo, karate, etc.) than for objectively measured sports, but we have no data with which to verify or quantify this asymmetry. The reduced-form estimate reported here should therefore not be read as identifying judge-bias per se, but rather as bounding the combined magnitude of the four candidate channels above.

Robustness to socioeconomic controls

Csurilla & Fertő (2023)[9] report that adding population and per-capita GDP controls to a country-level Olympic host regression attenuates the host coefficient by approximately 45%. Our Kokutai analog shows the *inverse*: adding population and GDP to the M1 host-only specification *increases* the absolute magnitude of the host coefficient by roughly a factor of 2.25 (for the ordered-rank outcome, from -6.11 to -13.73). This inversion is neither an error nor a paradox: it reflects that at the Kokutai the host is essentially a random draw from the size distribution of prefectures — the rotation schedule is fixed decades in advance and does not correlate with prefectural size — whereas Tokyo, an outlier on both population and GDP, dominates the non-host championship mass. Controlling for size therefore absorbs part of the non-host championship variance and (in relative terms) makes the host effect look larger. This mechanism is quantitatively confirmed by a Tokyo-exclusion sensitivity check (Table 4b): removing Tokyo from the panel ($n = 414$ vs. 423) purges all three non-host top-1 wins (2016, 2017, 2022 — all Tokyo) and produces complete separation on top-1, roughly doubling the M3 host coefficient (top-1 +9.09 → +17.66; ordered rank −13.73 → −24.00). The doubled coefficient should be read as directional evidence of the Tokyo-driven mechanism rather than as a preferred point estimate, since the separation regime distorts both the Fisher-information SE and the point estimate. This is a substantive finding: the Kokutai host bonus is *not* driven by prefectural size or economic capacity, and remains robust to standard socioeconomic controls.

Future work

A companion paper (in preparation) extends the categorical framework of the present study by introducing a continuous subjectivity gradient — scored on a Balmer et al. (2003)-consistent rubric across the full 40+ Kokutai sport catalogue with inter-rater reliability reported — and a sport-level exploratory screen for host-overperformance using newly collected competition-level top-N placement data. The companion paper is intended to complement (rather than supersede) the present categorical decomposition: the categorical contrast established here provides the qualitative direction, and the gradient/screen analyses would provide a finer-grained mechanistic decomposition together with case-study treatments of specific borderline sports (剣道 kendo, 柔道 judo, 相撲 sumo) that the current three-category framework treats uniformly.

Limitations

Ten limitations of the present study should be noted.

First, edition-level sport-by-prefecture score data are only publicly available for the 2024 Saga and 2025 Shiga editions. All earlier editions (through 2023 Kagoshima special) publish only prefecture-level overall standings, not the sport-by-prefecture breakdown. This restricts the Balmer et al. (2003)-style objective/subjective interaction test to a two-year cross-section (primary n = 4,744 obj-vs-subj cells; inclusive sensitivity n = 6,991 all cells) with only **2 treated prefecture clusters (Saga, Shiga) out of 47**. Under the few-treated-clusters diagnosis (Cameron & Miller 2015; MacKinnon & Webb 2017)[13,15], the cluster-robust variance formula is expected to be anti-conservative in this regime, and we explicitly report the wild-cluster bootstrap p-values (primary p = 0.175; inclusive sensitivity p = 0.156) alongside the cluster-robust p-values (primary p = 0.027; inclusive p = 0.031) in the Results section; the interaction magnitude should therefore be viewed as a first empirical anchor rather than a definitive magnitude. Year fixed effects in the cross-section are effectively a single dummy given the two-year window. If JSPO releases historical sport-by-prefecture breakdowns in the future, extending the interaction to a longer panel would substantially increase the treated-cluster count and tighten both the estimate and its inferential support.

Second, we could not obtain the JSPO judge-of-record roster (name + prefecture of affiliation) for any Kokutai edition. Four independent search paths — JSPO regulations catalog `tabid188`, seven individual-edition sampling, kendo-specific archives, and general web search — all returned negative. The Balmer et al. (2003)-style decomposition is therefore the only implicit test of judging bias we can offer; a direct judge-level Zitzewitz (2006)-style analysis[12] is not available for Kokutai data at present.

Third, the subjectively judged sports for which we could confirm high-fidelity edition-level score data (independent of the JSPO overall-standings pipeline) are essentially limited to kendo, which published a per-prefecture score table for the 2016 Iwate edition (71st) but discontinued the same table by 2022 Tochigi (77th). Judo, gymnastics, and other subjectively judged sports were not covered in this way. The 2024-2025 cross-section provides sport-level scores, but the pre-2024 subjective-sport history is patchy.

Fourth, the Suetsugu (2024, 2025) full texts are served through the Komazawa University institutional repository as WEKO3 octet-stream downloads that could not be reliably extracted; the citations are at bibliographic-plus-CiNii-link level only. This is a citation-quality limitation, not a data limitation, and does not affect the present study's numerical results.

Fifth, the Funahashi (2016) replication reported here omits Funahashi's population, GDP, headquarters, transfer-athlete, and participant controls because the ESRI 2008SNA prefectural GDP series (the reference socioeconomic dataset for our main-panel analysis) does not extend to fiscal 2002 or earlier. Our replicated coefficient of +1,575.45 accordingly represents a simpler specification than Funahashi's; the 5.9% gap versus Funahashi's +1,674.65 is well within what would be expected from the omitted controls, but a formal all-controls-included replication would require assembly of pre-2011 prefectural socioeconomic covariates from a separate ESRI historical release, which was outside the scope of the present study.

Sixth, the primary analysis panel window is 2012-2022 (calendar span; 9 non-cancelled editions; $n = 423$), constrained by the ESRI 令和4年度版 (2011-2022) coverage that is the newest release available as of manuscript preparation. When ESRI releases the complete 47-prefecture 令和5年度版 (extending through fiscal 2023; a partial release covering 31 prefectures and 2 designated cities is currently available), the panel can be extended by one year with essentially the same code path; when the 令和6年度版 releases (extending through 2024), the panel can be extended to include the 2024 Saga edition (currently reported only in the 2024-2025 cross-section). Extending the panel to 2024 would allow the 2024 Saga host-loss event to enter the main-panel analysis directly, tightening the event-study specification's identifying variation on the post-2016 cluster. All sample sizes reported in this manuscript ($n = 423$ main panel; $n = 4{,}744$ primary cross-section; $n = 6{,}991$ inclusive cross-section) are availability-driven — they reflect the maximum data currently released by JSPO and ESRI for each specification rather than a *ex ante* power-calculation design — so power/precision justifications are inherently tied to the JSPO/ESRI release schedule and would be strengthened by future releases of pre-2024 sport-by-prefecture breakdowns and post-2022 ESRI SNA series.

Seventh, we ran a Brant-style partial-proportional-odds diagnostic on the pooled ordered logit by fitting a separate binary logit at each rank threshold ($Y \leq j$ vs. $Y > j$, for $j = 1, \ldots, 8$) with prefecture-clustered standard errors and comparing the is_host coefficient across thresholds via an inverse-variance-weighted chi-square. Only the top-1 threshold ($Y \leq 1$) yielded a converged fit ($\beta = +9.09$, SE = 2.13, matching the top-1 pooled logit exactly); all higher thresholds ($Y \leq 2$ through $Y \leq 8$) hit numerical singularity because every host prefecture-year in the panel finishes in the top 8 (host_top8_rate = 9/9 = 100%), leaving no within-cell variance for those threshold-specific binary logits. A formal test of the proportional-odds assumption is therefore not tractable on this data-generating regime. Qualitatively, the same near-complete separation is itself circumstantial evidence that a single-slope representation is not violated by threshold heterogeneity — the host effect is so concentrated at the top that there is essentially no room for β_j to vary from threshold to threshold — but we cannot formally certify the assumption. We report the diagnostic honestly and acknowledge the limitation; the substantive conclusions in this paper rest on the pooled top-1 logit (which does not depend on the PO assumption at all), the pooled ordered logit (whose robustness we cannot formally test but which produces coefficients consistent with the top-1 fit), the Csurilla-style staged specification, and the 2024-2025 cross-section, and are therefore not overturned by the untestable status of the PO assumption in this particular ordered-logit specification.

Eighth, the β_{HS} (host × subjective interaction) coefficient in the 2024-2025 cross-section rests on an asymmetric identification structure that is worth making fully explicit alongside the few-treated-clusters caveat already noted in the First limitation above. In practical terms, the interaction is identified off the difference between Saga and Shiga in their subjective-vs-objective host-cell contrasts. Under the primary (semi-excluded) specification, roughly 38 subjective host cells (Saga 2024 subjective sports + Shiga 2025 subjective sports) carry essentially the entirety of the interaction identification, with the roughly 65 objective host cells serving as within-prefecture reference groups. Under the inclusive sensitivity specification, the same 38 subjective host cells contribute alongside 48 semi-subjective host cells and the 65 objective host cells; the primary specification's exclusion of semi-subjective cells from the reference base is precisely why the primary interaction point estimate (+20.27) exceeds the inclusive one (+16.68). This is not a generic "2 of 47" cross-section design; it is a specific 2-prefecture design in which the interaction's variance is a function of exactly how those two prefectures happened to differ in subjective-sport versus objective-sport performance during their respective host years. If Saga and Shiga had turned out to be substantively similar in their subjective-sport performance (a plausible counterfactual given both are mid-sized prefectures without an obviously dominant elite-athlete pipeline in judo/kendo/gymnastics), the interaction would be estimated off a small residual signal and the true SE would be correspondingly inflated. This asymmetric identification is one of the mechanical reasons the wild-cluster bootstrap p diverges so substantially from the naive cluster-robust p in both specifications (primary: 0.175 vs. 0.027; inclusive: 0.156 vs. 0.031) — the bootstrap re-samples over prefecture clusters and honestly reflects the fragility of a two-cluster identification base, whereas the cluster-robust SE leans on asymptotic approximations known to be anti-conservative in this regime (Cameron & Miller 2015; MacKinnon & Webb 2017)[13,15]. Extending the interaction test to a longer panel with more host prefectures would replace the current pairwise Saga-vs-Shiga structure with a broader within-prefecture identifying variance and correspondingly relax both the anti-conservative-SE concern and the two-prefecture asymmetric-identification concern.

Ninth, the two-layer event-study specification (moved to Supplementary Materials §S1 in the present manuscript) provides only thin identifying variation for the shock timing because the treated group (host prefectures) exhibits non-zero top-1 status essentially only in the host year itself; the event-study is therefore best interpreted as a design check on the structural-change characterization rather than as an independent identifying analysis of the shock timing. A continuous-score outcome would in principle allow a stronger event-study, but the 2003-2011 replication window is where continuous-score data are available and that window contains no post-2016 shocks. Extending the panel through 2024 (see Sixth limitation) would allow the 2024 Saga host-loss event to enter the main-panel analysis directly and would strengthen the identifying variation on the post-2016 cluster.

Tenth, the wild-cluster bootstrap reported in the primary and inclusive specifications (Table 5) and across all four classification variants (Table 5d) uses Rademacher weights (±1 with equal probability). With only two treated prefecture clusters ($G_1 = 2$), the discrete Rademacher weight distribution places non-trivial mass at just four possible sign patterns for the treated clusters (++, +-, -+, –), so the bootstrap p distribution may exhibit lumpiness at the tail-mass boundary. A robustness check with Mammen (1993) weights or an alternative continuous-weight family would sharpen the reported p-values, but is unlikely to change the qualitative conclusion (all reported bootstrap p-values fall in the 0.15-0.18 range, well above conventional 0.05 significance); we defer this robustness check to future work as data availability from additional host prefectures accrues.

Conclusion

Japan's National Sports Festival shows a host bonus that appears directionally non-uniform across sport types. The results are directionally consistent with a larger host bonus in subjectively judged sports than in objectively measured sports under the primary obj-vs-subj-pure specification (host × subjective interaction = +20.27 points on a host main effect of +11.18 points, i.e., point estimate roughly 3× the objective bonus; cluster-robust $p = 0.027$; wild-cluster bootstrap $p = 0.175$ under the few-treated-clusters correction on 2 treated prefectures out of 47), with a smaller (roughly 2×) point estimate under the inclusive sensitivity specification retaining all sports (+16.68 on +16.60; cluster-robust $p = 0.031$, bootstrap $p = 0.156$), directionally matching the Balmer, Nevill & Williams (2003) decomposition established at the summer Olympics. The overall host bonus is robust to prefecture-level population and GDP controls (indeed strengthened by them, unlike at the Olympics), replicates Funahashi et al. (2016)'s 2003-2011 headline coefficient within 5.9%, and survives an extension of Funahashi's panel through 2022. The structural change in host championship rate around 2016 is best characterized as two clusters (2016-17 and 2022-24) separated by a return to host wins in 2018-19, not a single break. The finding closes the specific quantitative gap left by Funahashi (2016) and provides Japanese panel-econometric evidence complementary to the qualitative critique of Suetsugu (2024, 2025). Whether the observed interaction reflects judging bias, subject-sport-specific crowd effects, or subject-sport-specific practice-environment effects cannot be uniquely identified from the present design; the interaction is directionally supported by the descriptive monotonicity, the log-outcome robustness check, and a three-variant classification sensitivity in which all four variants preserved positive point estimates (range +18.99 to +31.81; Table 5d). The combat-excluded variant's cluster-robust $p < 0.001$ shares the same two-treated-cluster limitation as the primary specification — the wild-cluster bootstrap p (0.174) is essentially identical to the primary's (0.175), so its inferential status under the few-treated-clusters correction is the same as the primary — and should be read in the same "first empirical anchor" register that longer panels could sharpen into a decisive test.

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bibliographic verification records archived with the analysis code document the verification process, including two corrections to entries that appeared in earlier drafts (Funahashi et al. 2016 co-author list and journal; Csurilla & Fertő 2023 author count).

Author Contributions (Contributor Roles Taxonomy, CRediT)

Mizuki Shirai: Conceptualization, Methodology, Investigation, Data Curation, Formal Analysis, Software, Writing — Original Draft, Writing — Review & Editing, Visualization, Project Administration.

Conflict of Interest

The author declares no conflicts of interest per International Committee of Medical Journal Editors (ICMJE) guidelines.

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Data Availability

All input data used in this study are publicly available. Prefecture rankings are from the Nagano Prefecture Sports Association (https://www.nagano-sports.or.jp/kokutai/record/high_rank.html); individual-edition PDFs and 2024-2025 Excel breakdowns are from the JSPO official archive (https://www.japan-sports.or.jp/kokutai/tabid183.html); socioeconomic covariates are from the Cabinet Office ESRI prefectural accounts (https://www.esri.cao.go.jp/jp/sna/data/data_list/kenmin/files/contents/main_2022.html).

Analysis code (Python 3.14, pandas 3.0.5, statsmodels 0.14.6, matplotlib 3.11.1, weasyprint 69.0) — including the data-assembly scripts, all regression modules (pooled ordered/binary logit, Csurilla-style staged specification, Funahashi replication OLS, 2024-2025 cross-section), the wild-cluster bootstrap and Brant-style partial-proportional-odds diagnostic implementations, the plotting pipeline, and the PDF build pipeline — will be released to a public GitHub repository and permanently archived on Zenodo (with a DOI) concurrent with SSRN posting. This commitment is made explicit to avoid an asymmetric-disclosure gap in which AI-usage transparency (see Acknowledgments — LLM-assisted development is fully disclosed at the point of first posting) would run ahead of code transparency and thereby impair independent reviewer verification of the LLM-assisted numerical results reported here.

Tables

Table 1. Descriptive statistics for the 2012-2022 emperor's-cup panel

Statistic	Value
Number of prefecture-year observations	423
Number of host-year cells	9
Number of non-host prefecture-year cells	414
Host cells that finished in top 1 (championship)	6 (66.7%)
Host cells that finished in top 8	9 (100%)
Non-host cells that finished in top 1	3 (0.72%)
Non-host cells that finished in top 8	62 (15.0%)
Host mean rank when ranked (top 8)	1.33

The 9 host years in the panel window (2012-2022) are 2012 Gifu, 2013 Tokyo, 2014 Nagasaki, 2015 Wakayama, 2016 Iwate, 2017 Ehime, 2018 Fukui, 2019 Ibaraki, and 2022 Tochigi (n = 9 non-cancelled non-special host years; 2020 Kagoshima and 2021 Mie were cancelled for COVID-19, and 2023 Kagoshima special and 2024 Saga fall outside the 2012-2022 window). Of the 9 host years, 6 were host victories (2012 Gifu, 2013 Tokyo self-host, 2014 Nagasaki, 2015 Wakayama, 2018 Fukui, 2019 Ibaraki) and 3 were host losses (2016 Iwate, 2017 Ehime, 2022 Tochigi — all won by Tokyo). The 2013 Tokyo cell is retained in the regression sample (n = 423) and is included in the "6 of 9" descriptive rate reported in Results: Tokyo's host-year 1st-place finish (host_win = 1) is a legitimate host-win observation. As a sensitivity check, excluding 2013 Tokyo yields 5 of 8 = 62.5%, still two orders of magnitude above the non-host per-year top-1 rate of 0.72% and in the same direction as the primary "6 of 9" rate — confirming that the pattern is not an artifact of Tokyo's generalized non-host dominance.

Table 2. Main-panel regression results (2012-2022 tennou, n = 423)

Specification	Coefficient (host)	SE	95% CI	p-value	log-likelihood	McFadden pseudo-R^2	Notes
Ordered logit, pooled (+ log_pop + log_gdp)	-13.73	2.03	[-17.71, -9.75]	< 0.001	-173.85	0.487	Negative = better rank
Ordered logit, pref FE + year FE (+ log_pop + log_gdp)	-49.56	172.45	[-387.56, +288.43]	0.77	-111.19	0.672	Separation-driven degeneracy
Top-1 logit, pooled (+ log_pop + log_gdp)	+9.09	2.13	[+4.92, +13.27]	< 0.001 (exact 1.9×10^{-5})	-12.75	0.707	
Top-1 logit, pref FE + year FE (+ log_pop + log_gdp)	+1,877.3	(undefined)	(undefined)	(undefined)	-0.00	1.000	Complete separation (pseudo-R^2 = 1 is a signature, not a fit)

Prefecture-clustered standard errors (47 clusters, Liang-Zeger sandwich) for the pooled specifications; Fisher-information for the fixed-effect specifications, where prefecture FE renders further clustering mechanically redundant and where SE degeneracy is itself a diagnostic signature of complete separation. All specifications above include log-population and log-GDP as covariates by construction; the "pooled" label denotes the absence of prefecture and year fixed effects, not the absence of covariates. The two pooled rows above are numerically identical (coefficient and clustered SE) to the M3 (+ log GDP) row of Table 4. Fixed-effect specifications are reported for transparency but not interpreted as coefficient estimates; see Methods.

Table 3. Funahashi (2016) replication (2003-2011 tennou, n = 423)

Specification	Coefficient (host)	SE (pref clusters)	p-value	R^2
Base (pref FE + year FE, no controls)	+1,575.45	57.58	< 0.001	0.94
Pooled OLS, no fixed effects	+1,733.29	138.26	< 0.001	0.33
Extended (2003-2012, +1 year)	+1,604.36	60.08	< 0.001	0.94
Funahashi (2016) reported base	+1,674.65	—	(n.r.)	0.86

47 prefecture clusters. Our replication omits Funahashi's socioeconomic controls (see Limitations); the 5.9% gap versus Funahashi's headline coefficient is within the expected range for the omitted-controls difference.

Table 4. Csurilla-style staged specification (2012-2022 tennou, n = 423)

Stage	Model	Host coefficient (top-1 / rank)	SE (top-1 / rank)	95% CI (top-1 / rank)	p (top-1 / rank)	log-likelihood (top-1 / rank)	pseudo-R² (top-1 / rank)
M1	Host only	+5.61 / -6.11	1.09 / 1.21	[+3.47, +7.75] / [-8.48, -3.74]	< 0.001 / < 0.001	-23.50 / -308.20	0.461 / 0.091
M2	+ log population	+10.90 / -13.17	3.35 / 2.01	[+4.33, +17.47] / [-17.11, -9.23]	0.0011 / < 0.001	-14.83 / -176.49	0.660 / 0.479
M3	+ log GDP	+9.09 / -13.73	2.13 / 2.03	[+4.92, +13.27] / [-17.71, -9.75]	< 0.001 / < 0.001	-12.75 / -173.85	0.707 / 0.487
M4	+ prefecture FE	+498 / -116	NaN / NaN	(undefined) / (undefined)	(degenerate) / (degenerate)	-3.43 / -112.51	0.921 / 0.668
M5	+ year FE	+1,877 / -49.56	NaN / 172.45	(undefined) / [-387.56, +288.43]	(degenerate) / 0.77	-0.00 / -111.19	1.000 / 0.672

Standard errors are prefecture-clustered (47 clusters, Liang-Zeger sandwich) for M1-M3 and Fisher-information for M4/M5 (where prefecture FE renders clustering mechanically redundant and complete separation drives SEs to degeneracy). The M3 row is numerically identical (coefficient and clustered SE) to the corresponding pooled row of Table 2. The attenuation Csurilla & Fertő (2023) documented at the Olympics (host coefficient shrinks by ~45% from M1 to fully adjusted) is inverted at the Kokutai: M1 → M3 raises the top-1 coefficient by ~62% and the ordered-rank coefficient by ~125%. See Discussion for interpretation.

Table 4b. Csurilla-style staged specification with Tokyo excluded (Tokyo-exclusion sensitivity)

Stage	Model	Host coefficient (top-1 / rank)	SE (top-1 / rank)	p (top-1 / rank)	log-likelihood (top-1 / rank)	pseudo-R² (top-1 / rank)
M1	Host only	+14.61 / -18.27	0.75 / 1.14	< 0.001 / < 0.001	-5.29 / -264.36	0.804 / 0.124
M2	+ log population	+17.40 / -37.19	0.78 / 1.68	< 0.001 / < 0.001	-5.29 / -162.47	0.805 / 0.461
M3	+ log GDP	+17.66 / -24.00	0.86 / 1.98	< 0.001 / < 0.001	-5.15 / -162.46	0.810 / 0.461
M4	+ prefecture FE	+122.8 / -98.0	NaN / NaN	(degenerate) / (degenerate)	-0.00 / -107.00	1.000 / 0.645
M5	+ year FE	+67.3 / -68.3	NaN / NaN	(degenerate) / (degenerate)	-0.00 / -105.81	1.000 / 0.649

n = 414 (= 423 total − 9 Tokyo prefecture-year cells; equivalent to 46 prefectures × 9 years, with 2013 Tokyo's own self-host year also removed so the effective host count drops from 9 to 8). Prefecture-

clustered SE on 46 clusters for M1–M3; Fisher-information for M4/M5. All three non-host top-1 wins in the full-sample panel (2016, 2017, 2022) are Tokyo wins and are removed by the exclusion, producing a complete separation on top-1 (8 of 8 remaining hosts won; 0 of 406 non-hosts won). The near-doubling of the M3 host coefficient (top-1 +9.09 → +17.66; ordered rank −13.73 → −24.00 vs. Table 4) is the direct quantitative confirmation of the Csurilla-reverse mechanism discussed in Results and Discussion — Tokyo's non-host championship mass was suppressing the estimated host coefficient in the full sample. This is offered as a mechanism diagnostic, not as a preferred point estimate; the separation regime drives Fisher-information SEs downward across all rows and inflates the coefficient beyond the full-sample identifying variation (M4/M5 rows in particular), so the M1-M3 SE cells should be read as separation-regime diagnostics rather than as inference-eligible standard errors even though the maximum-likelihood optimizer returned finite values on early convergence; Table 4 (with Tokyo retained) remains the primary quantitative claim.

Table 5. 2024-2025 cross-section: Balmer et al. (2003) host × subjective interaction (primary n = 4,744 obj-vs-subj cells; inclusive sensitivity n = 6,991 all cells; 2 treated prefectures out of 47)

Specification	Sample	Coefficient (host)	Coefficient (host × subj)	SE (int)	95% CI (int)	p (int)	Cluster SE
Primary: obj vs subj pure (semi excluded), cluster-robust p	n = 4,744 (28 sports: obj + subj only)	+11.18	**+20.27**	9.15	[+2.33, +38.20]	**0.027**	47 pref clusters
Primary: obj vs subj pure (semi excluded), wild-cluster bootstrap (Rademacher, B = 999, restricted null)	n = 4,744	+11.18	+20.27	9.15	**[-0.82, +41.36]** (pivotal 95%)	**0.175**	47 pref clusters (2 treated)
Exploratory diagnostic — cluster SE degenerate: Zero-imputed sensitivity (106 absent cells = 0; non-participation vs missing-data check; post-hoc reviewer-driven), obj vs subj pure	n = 4,841	+11.32	**+19.86** (β_HS)	(degenerate)	(cluster SE numerically undefined)	(cluster SE numerically undefined)	—
Sensitivity: inclusive (all sports incl. semi-subjective/team), cluster-robust p	n = 6,991 (41 sports)	+16.60	+16.68	7.72	[+1.55, +31.81]	0.031	47 pref clusters
Sensitivity: inclusive, wild-cluster bootstrap (Rademacher, B = 999, restricted null)	n = 6,991	+16.60	+16.68	7.72	**[-0.42, +33.86]** (pivotal 95%)	0.156	47 pref clusters (2 treated)
Diagnostic only — cluster SE not identified: Three-way (obj/semi/subj interactions), inclusive	n = 6,991	+12.99	+20.31 (β_HS), +8.51 (β_H,semi)	NaN	(cluster SE numerically undefined)	(cluster SE numerically undefined)	—
Diagnostic only — cluster SE not identified: Primary + sport × trophy interaction (tennou-vs-kougou within-sport	n = 4,744	+9.71	**+21.97** (β_HS)	NaN	(cluster SE numerically undefined)	(cluster SE numerically undefined)	—

Specification	Sample	Coefficient (host)	Coefficient (host × subj)	SE (int)	95% CI (int)	p (int)	Cluster SE
diagnostic), obj vs subj pure							
Log(1 + score) inclusive, cluster-robust p	n = 6,991	+0.467	+0.293	0.170	[-0.040, +0.626] (crosses 0)	0.084 (marginal)	47 pref clusters

Primary specification (top two rows) excludes the 13 semi-subjective/team sports (soccer, basketball, volleyball, handball, hockey, rugby, softball, baseball, badminton, table tennis, soft tennis, tennis, ice hockey) to isolate a pure obj vs subj comparison; this restores the intended semantic content of β_HS (subjective vs objectively measured reference) that the inclusive specification blends with a semi-subjective mass. Inclusive specification (middle two rows) retains all sports as sensitivity. Both cluster-robust interaction p-values fall below 0.05 and both wild-cluster bootstrap p-values fall in the 0.15-0.18 range under the few-treated-clusters correction (Cameron, Gelbach & Miller 2008; Cameron & Miller 2015), so the qualitative reading (significant under naive test, non-significant under bootstrap) is invariant to semi-exclusion. The primary specification yields a larger interaction point estimate (+20.27) than the inclusive (+16.68), consistent with the mixing bias the primary spec is designed to eliminate. Three diagnostic rows (three-way, primary + sport × trophy interaction, and zero-imputed sensitivity) report point estimates only owing to distinct cluster-meat-matrix pathologies: few-clusters regime for the first two (three-way: k=94 / G=47 ≈ 2; sport × trophy: k=131 / G=47 ≈ 2.8; Kezdi 2004) and augmented-cartesian rank deficiency for the zero-imputed row (see Methods §3). All three diagnostic β_HS point estimates (+20.31, +21.97, +19.86) fall within 9% of the primary specification's +20.27, corroborating direction and magnitude across five internally consistent estimates. Prefecture, sport, year, and trophy fixed effects included in all inference-eligible specifications; 47 prefecture clusters (Saga and Shiga are the two treated clusters), robust SE throughout.

Table 5d. Classification-variant sensitivity for primary specification

Variant	Subjective sports (n)	n obs	β_H	β_HS	SE (int)	cluster-robust p	wild-cluster bootstrap p
Default (primary reference; Balmer et al. (2003))	11 (incl. combat)	4,744	+11.18	**+20.27**	9.15	0.027	0.175
Pure judged (subj = 体操 + 空手道 + なぎなた + 馬術)	4	4,744	+15.70	+18.99	17.71	0.284	0.647
Combat sports excluded (subj = 体操 + 馬術 only)	2	3,359	+10.71	**+31.81**	8.20	**< 0.001**	**0.174**
Combat-to-semi (subj = 8 excl. fencing/wrestling/sumo)	8	4,283	+10.30	+26.40	11.01	0.016	0.153

Sensitivity of the primary specification to the delimitation of the "subjective" category, with wild-cluster bootstrap p-values (Rademacher weights, $B = 999$, restricted null, seed matched to Table 5 primary bootstrap) reported consistently across all four variants (closing the "selective standard" gap in which Table 5 primary/inclusive rows had bootstrap correction while Table 5d variants did not). All four variants preserve $\beta\_HS > 0$ (range +18.99 to +31.81); the primary reference (+20.27) lies inside this range. The combat-excluded variant, which isolates pure artistic scoring (gymnastics + equestrian only), yields the largest cluster-robust point estimate (+31.81, cluster-robust $p < 0.001$), consistent with a mechanism in which host bias is strongest under pure artistic scoring rather than under combat-referee judgment. Under the wild-cluster bootstrap correction applied to all four variants, however, the no-combat variant's bootstrap p (0.174) is essentially identical to the primary specification's bootstrap p (0.175; Table 5) — its inferential status under the few-treated-clusters correction is the same as the primary, not uniquely stronger; the cluster-robust $p < 0.001$ is anti-conservative in the same 2-treated-prefecture regime that produced the primary's $p = 0.027 \rightarrow 0.175$ divergence. The pure-judged variant's larger SE (17.71) and bootstrap p (0.647) both reflect the reduced subjective-host cell count (n_subj_host ≈ 14). All variants estimated on the primary spec (semi-excluded, prefecture + sport + year + trophy fixed effects, prefecture-clustered SE on 47 clusters).

Table 6. Descriptive host-boost by sport category (2024-2025, per-sport score)

Category	Mean non-host	Mean host	Cells (non-host)	Cells (host)	Host boost (points)
Objective (n = 17 sports)	26.03	43.63	2,920	65	**+17.60**
Semi-subjective / team (n = 13 sports)	20.85	47.03	2,199	48	**+26.18**
Subjective (n = 11 sports)	19.33	57.24	1,721	38	**+37.91**

Monotonic increase in raw host boost across the three categories matches the direction predicted by Balmer et al. (2003). The subjective category's +37.9 is the largest of the three and is roughly 2.15× the objective category's +17.6, consistent with the model-implied ratios in Table 5 (primary: (+11.18 + +20.27) / +11.18 ≈ 2.81×; inclusive: (+16.60 + +16.68) / +16.60 ≈ 2.00×).

Table 7. Cross-national context: host-scoring institutional arrangements

System	Host score bonus	Empirical host effect	Reference
Japan Kokutai (present study)	None (no institutional bonus in JSPO scoring)	+31 pts subj / +11 pts obj per sport (primary spec; +33/+17 under inclusive sensitivity; see Table 5)	Present study
Korea national comprehensive sports	20% score bonus (10% from 2001, 20% from 2010)	Institutionally amplified	Kim (2025), Munhwa Ilbo[8]
China National Games	(institutional bonus not documented in accessible sources)	Qualitatively described (5 characteristics: high efficacy, gradual increase, delayed effect, exclusivity, double-edgedness)	Lan & Yu (2012)[11]
Summer Olympics	None	Attenuates ~45% under population + GDP + country FE	Csurilla & Fertő (2023)[9]

The Japanese case is the only one of the four in which a large host effect emerges without any institutional score bonus and without an obvious socioeconomic driver.

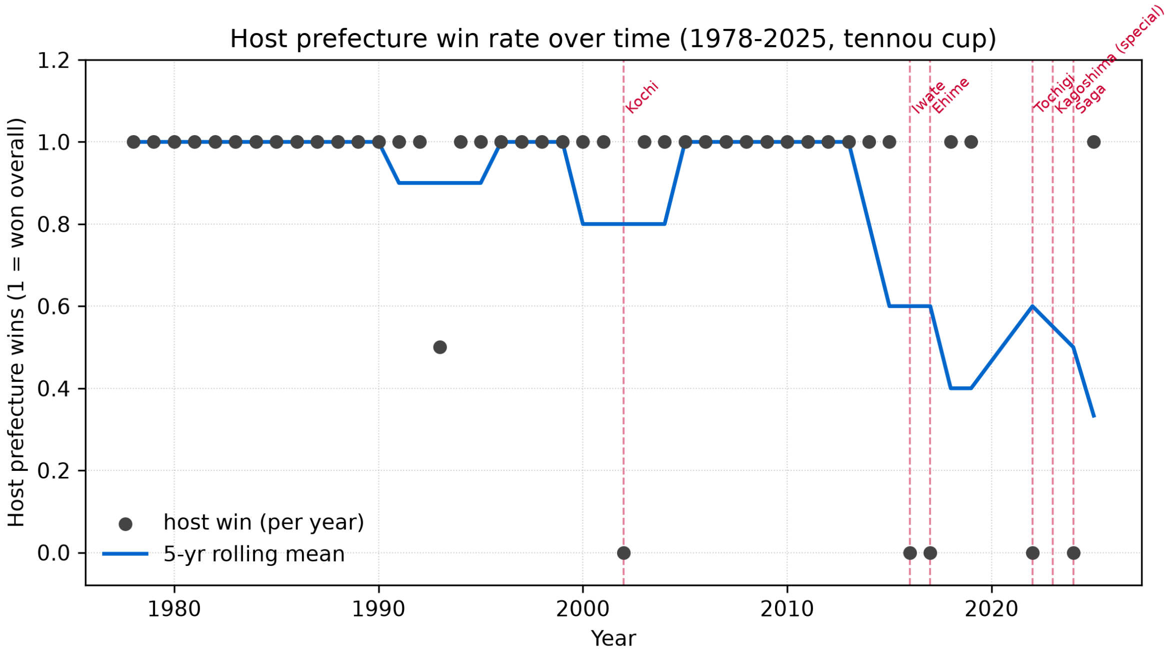


Figure 1. Host-championship rate at the Kokutai emperor's cup, 1978-2025. Each point is one edition (with the 2020/2021 cancelled and 2023 special-edition years handled as noted in the figure). Six host-loss years (2002 Kochi, 2016 Iwate, 2017 Ehime, 2022 Tochigi, 2023 Kagoshima special, 2024 Saga) are highlighted in red. The two post-2016 host-loss clusters (2016-17 and 2022-24) are separated by host wins in 2018 Fukui and 2019 Ibaraki. *Note on the 1993 point at y ≈ 0.5.* The 48th edition (1993) was co-hosted by Kagawa and Tokushima — the only jointly hosted edition in the 1948-2025 series. Kagawa finished 1st (host_win = 1) and Tokushima finished 2nd (host_win = 0), so the two-prefecture host mean is (1 + 0) / 2 = 0.5. This is a historical feature of the schedule, not a data artifact.

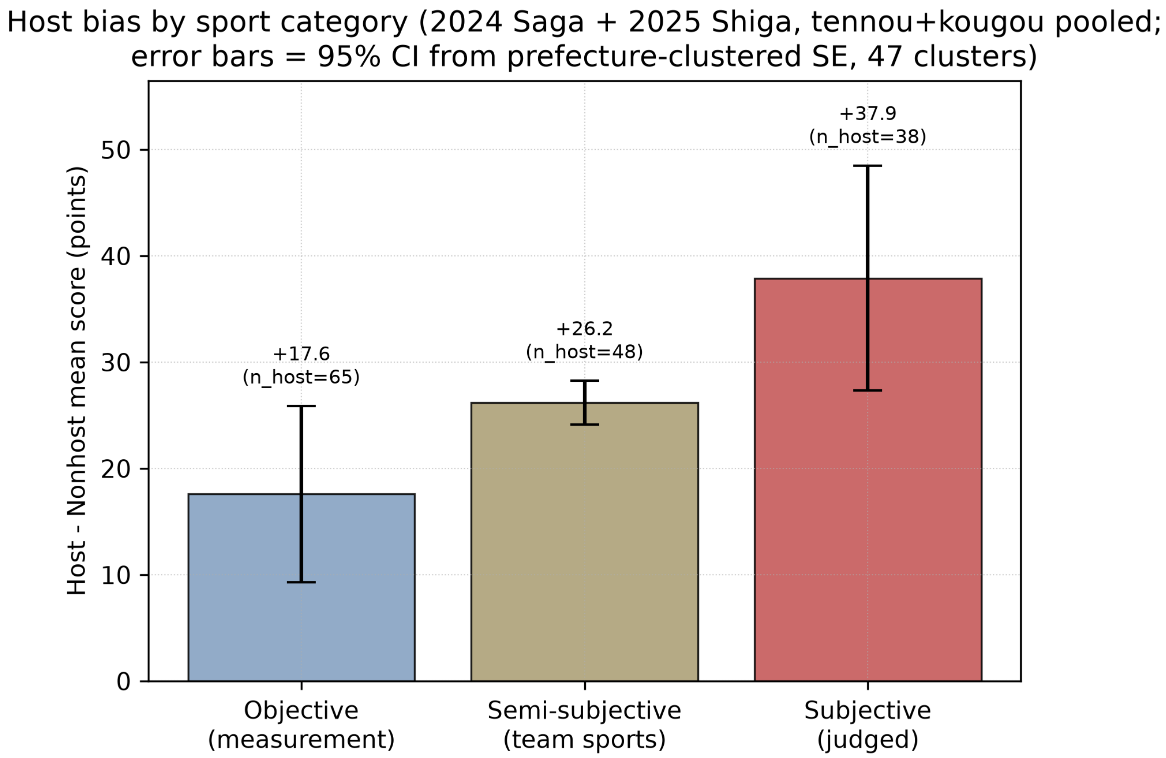


Figure 2. Host-versus-non-host per-sport score gap, by Balmer et al. (2003) sport category, in the 2024-2025 stacked cross-section (n = 6,991). Bars show the mean host boost in points; error bars show 95% confidence intervals (1.96 × prefecture-clustered SE, 47 clusters, per-category `score ~ is_host` OLS). Objective sports (n = 17, n_host_cells = 65): +17.6 points. Semi-subjective / team sports (n = 13, n_host_cells = 48): +26.2 points. Subjective sports (n = 11, n_host_cells = 38): +37.9 points. The monotonic ordering matches the Balmer et al. (2003) prediction.

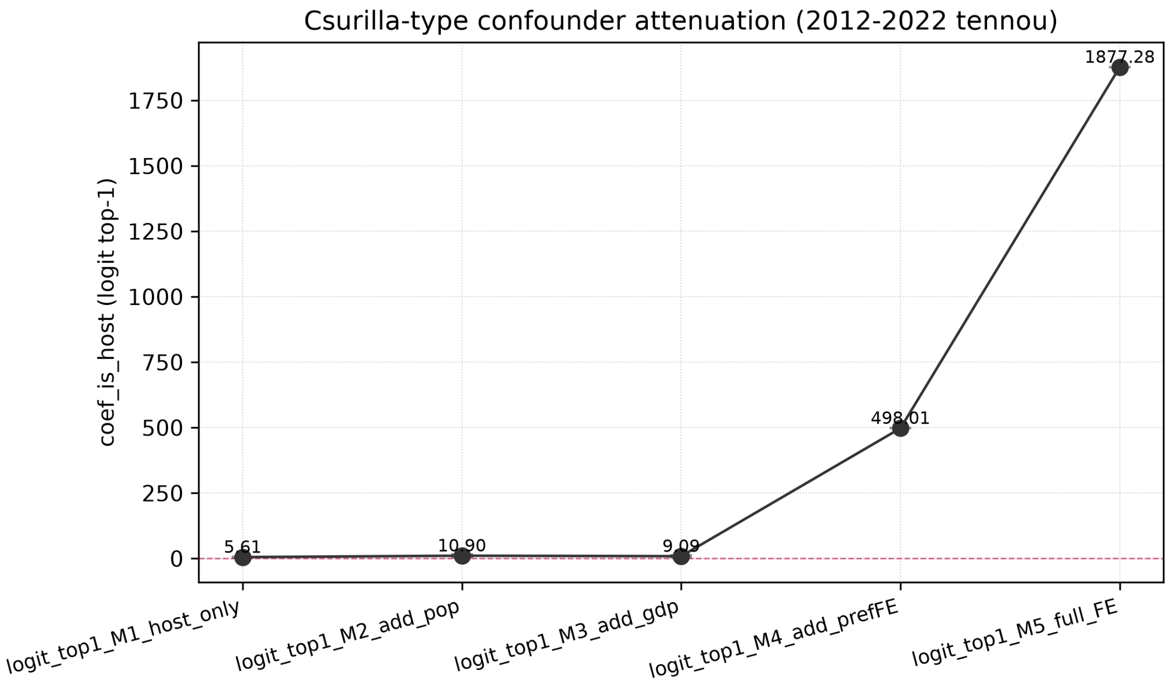


Figure 4. Csurilla-style M1→M5 staged specification, top-1 outcome. The host coefficient grows in absolute magnitude as population and GDP controls are added (M1 = +5.61, M2 = +10.90, M3 = +9.09), the opposite of the ~45% attenuation Csurilla & Fertő (2023) report for the Olympics. M4 and M5 exhibit separation-driven degeneracy and are shown truncated.

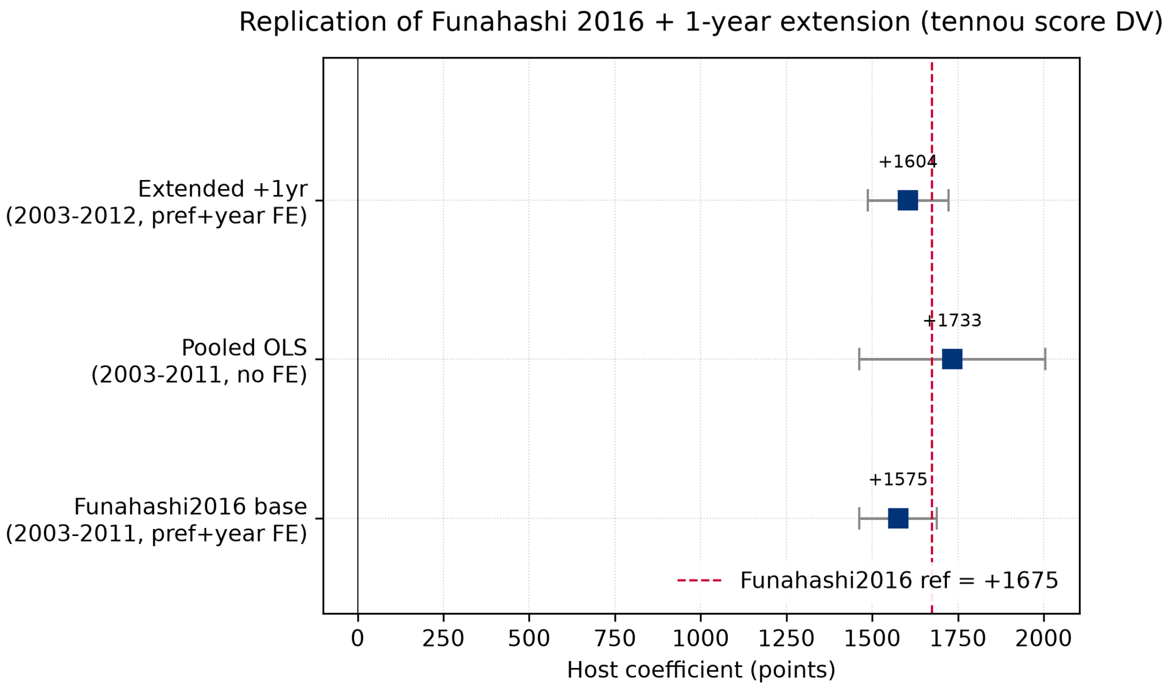
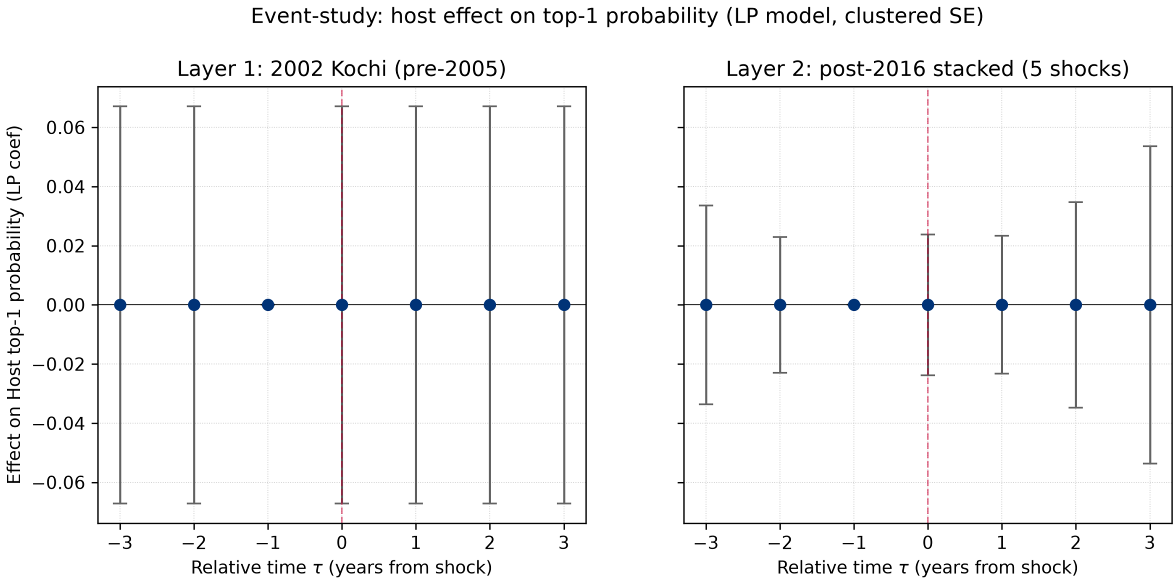


Figure 5. Funahashi (2016) replication, three specifications on the 2003-2011 emperor's-cup total-score panel. `funahashi_base` (prefecture FE + year FE, no controls) recovers +1,575.45; `pooled_no_fe` gives +1,733.29; `extended_2003_2012` (+1 year) gives +1,604.36. Funahashi's headline coefficient of +1,674.65 is shown as a dashed reference line; error bars are 95% confidence intervals (1.96 × SE). The `pooled_no_fe` specification lies within one SE of the reference; `funahashi_base` and `extended_2003_2012` lie within 1.96 SE (95% CI) of the reference.

Supplementary Figures



Supplementary Figure S1. Two-layer event-study τ-coefficient trails on top-1 (championship) status. Layer 1: 2002 Kochi single host-loss event (pre-2005 *furusato*-athlete-rule regime). Layer 2: post-2016 stacked five host-loss events (2016 Iwate, 2017 Ehime, 2022 Tochigi, 2023 Kagoshima special, 2024 Saga). The identifying variation is thin (see supplementary text above); the figure is included as a design check rather than as an independent identifying analysis.